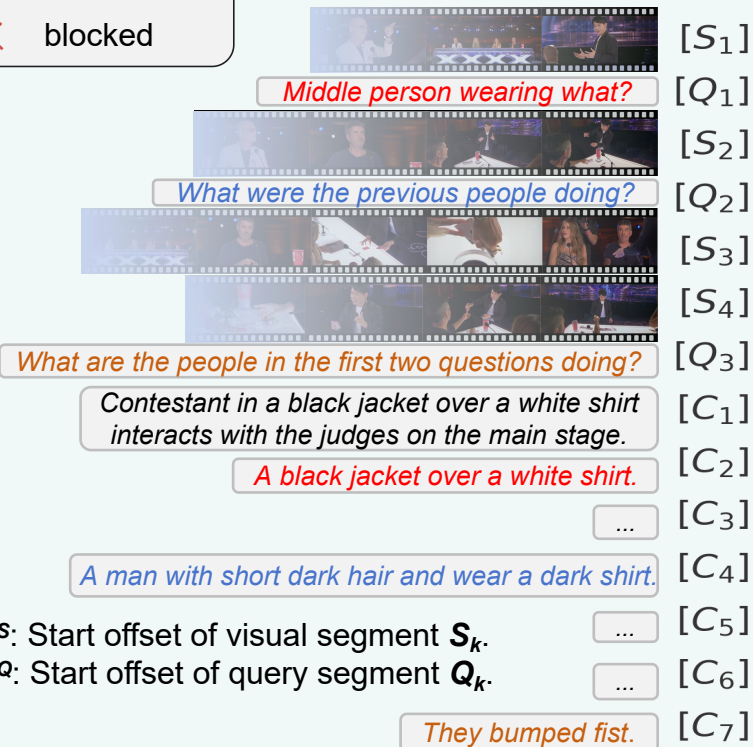


(a) Segment-level Streaming Mask and Streaming Positional Encoding

○ attendable
✗ blocked

[S₁] [Q₁] [S₂] [Q₂] [S₃] [S₄] [Q₃] [C₁] [C₂] [C₃] [C₄] [C₅] [C₆] [C₇]



	[S ₁]	[Q ₁]	[S ₂]	[Q ₂]	[S ₃]	[S ₄]	[Q ₃]	[C ₁]	[C ₂]	[C ₃]	[C ₄]	[C ₅]	[C ₆]	[C ₇]
[S ₁]	○	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
[Q ₁]	○	○	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
[S ₂]	○	○	○	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
[Q ₂]	○	○	○	○	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
[S ₃]	○	○	○	○	○	✗	✗	✗	✗	✗	✗	✗	✗	✗
[S ₄]	○	○	○	○	○	○	✗	✗	✗	✗	✗	✗	✗	✗
[Q ₃]	○	○	○	○	○	○	○	✗	✗	✗	✗	✗	✗	✗
[C ₁]	○	✗	✗	✗	✗	✗	✗	○	✗	✗	✗	✗	✗	✗
[C ₂]	○	○	✗	✗	✗	✗	✗	○	○	✗	✗	✗	✗	✗
[C ₃]	○	○	○	✗	✗	✗	✗	○	○	○	✗	✗	✗	✗
[C ₄]	○	○	○	○	✗	✗	✗	○	○	○	○	✗	✗	✗
[C ₅]	○	○	○	○	○	✗	✗	○	○	○	○	○	✗	✗
[C ₆]	○	○	○	○	○	○	✗	○	○	○	○	○	○	✗
[C ₇]	○	○	○	○	○	○	○	○	○	○	○	○	○	○

- B_k^S : Start offset of visual segment S_k .
- B_k^Q : Start offset of query segment Q_k .

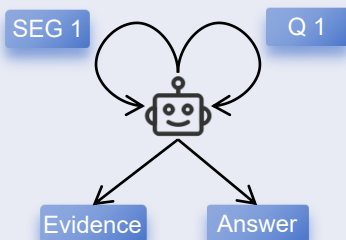
They bumped fist.

- B_k^C : Start offset of CoT segment C_k .

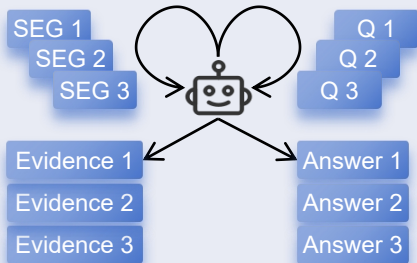
$B_1^S(0)$ B_1^Q B_2^S B_2^Q B_3^S B_4^S B_3^Q $B_1^C(0)$ B_2^C B_3^C B_4^C B_5^C B_6^C B_7^C

(b) Three-stage training strategy

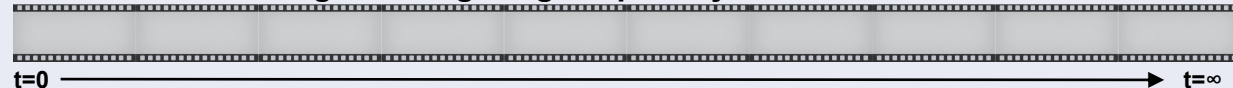
Stage 1: Single-round CoT



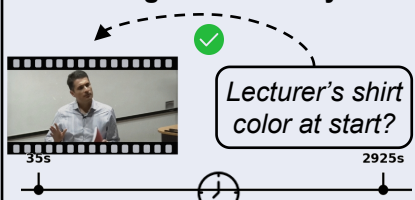
Stage 2: Multi-round CoT



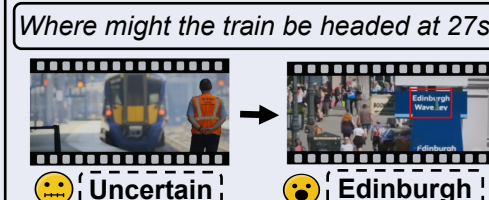
Stage 3: Long-range capability



Long-term memory



Uncertainty handling



Distractor Segment learning

